Pandas - Read CSV Notes

Handwritten-Style Study Notes

What is Pandas?

Pandas is an open-source Python library used for data manipulation, cleaning, analysis, and working with tabular data efficiently.

Main Uses

• Data Handling – Manage missing values
• Data Cleaning – Remove duplicates and inconsistencies
• Data Transformation – Filter, sort, and modify data
• Data Visualization – Represent data graphically

Data Structures in Pandas

1. Series

A one-dimensional labeled array capable of storing any data type.

2. DataFrame

A two-dimensional tabular data structure consisting of rows and columns.

Important read_csv() Parameters

usecols → Read only selected columns.
index_col → Set a column as row labels.
na_values → Convert custom values to NaN.
sep → Specify custom delimiters.
nrows → Read only a fixed number of rows.
skiprows → Skip selected rows while reading.
parse_dates → Convert columns into datetime objects.

Reading CSV from URL

Pandas can directly load CSV files from a web URL using read_csv(url).

Quick Revision

- Series = 1D structure
- DataFrame = 2D structure
- usecols = Select columns
- index_col = Set index
- na_values = Handle missing data
- parse_dates = Date conversion
- nrows / skiprows = Control rows loaded

Practice Questions

1. Difference between Series and DataFrame?
2. Why use usecols?
3. What is index_col?
4. When should na_values be used?
5. Difference between nrows and skiprows?
6. Why is parse_dates important?